

Networking Multiple Choice Questions

1. An IPv4 address consists of _____ bits.

- A) 4
- B) 8
- C) 32
- D) 64

2. In IPv4, class _____ has the greatest number of addresses in each block.

- A) A
- B) B
- C) C
- D) D

3. Identify the class of the following IPv4 address: 4.5.6.7.

- A) A
- B) B
- C) C
- D) none of the above

4. Identify the class of the following IPv4 address: 229.1.2.3.

- A) A
- B) B
- C) D
- D) none of the above

5. Identify the class of the following IPv4 address: 191.1.2.3.

- A) A
- B) B
- C) C
- D) none of the above

6. What is the result of ANDing 255 and 15?

- A) 255
- B) 15

- C) 0
- D) none of the above

7. What is the result of ANDing 0 and 15?

- A) 255
- B) 15
- C) 0
- D) none of the above

8. What is the result of ANDing 254 and 15?

- A) 254
- B) 14
- C) 0
- D) none of the above

9. What is the result of ANDing 192 and 65?

- A) 192
- B) 65
- C) 64
- D) none of the above

10. Which one is not a contiguous mask?

- A) 255.255.255.254
- B) 255.255.224.0
- C) 255.148.0.0
- D) all are

11. The number of addresses in a class C block is ____.

- A) 65,534
- B) 16,777,216
- C) 256
- D) none of the above

12. The number of addresses in a class B block is ____.

- A) 65,536
- B) 16,777,216

- C) 256
- D) none of the above

13. The number of addresses in a class A block is _____.

- A) 65,534
- B) 16,777,216
- C) 256
- D) none of the above

14. The number of addresses assigned to an organization in classless addressing _____.

- A) can be any number
- B) must be a multiple of 256
- C) must be a power of 2
- D) none of the above

15. The first address assigned to an organization in classless addressing _____.

- A) must be a power of 4
- B) must be evenly divisible by the number of addresses
- C) must belong to one of the A, B, or C classes
- D) none of the above

16. Which address could be the beginning address of a block of 32 classless addresses?

- A) 2.4.6.5
- B) 2.4.6.16
- C) 2.4.6.64
- D) none of the above

17. Which address could be the beginning address of a block of 16 classless addresses?

- A) 2.4.6.5
- B) 2.4.6.15
- C) 2.4.6.62
- D) none of the above

18. Which address could be the beginning address of a block of 256 classless addresses?

- A) 2.4.6.5
- B) 2.4.6.15
- C) 2.4.6.0
- D) none of the above

19. What is the first address of a block of classless addresses if one of the addresses is 12.2.2.76/27?

- A) 12.2.2.0
- B) 12.2.2.32
- C) 12.2.2.64
- D) none of the above

20. What is the first address of a block of classless addresses if one of the addresses is 12.2.2.76/10?

- A) 12.0.0.0
- B) 12.2.0.0
- C) 12.2.2.2
- D) none of the above

21. What is the first address of a block of classless addresses if one of the addresses is 12.2.2.127/28?

- A) 12.2.2.0
- B) 12.2.2.96
- C) 12.2.2.112
- D) none of the above

22. Find the number of addresses in a block of classless addresses if one of the addresses is 12.2.2.7/24.

- A) 32
- B) 64
- C) 256
- D) none of the above

23. Find the number of addresses in a block of classless addresses if one of the addresses is 12.2.2.7/30.

- A) 2
- B) 4
- C) 8
- D) none of the above

24. What is the last address of a block of classless addresses if one of the addresses is 12.2.2.127/28?

- A) 12.2.2.16
- B) 12.2.2.112
- C) 12.2.2.127
- D) none of the above

25. What is the last address of a block of classless addresses if one of the addresses is 12.2.2.6/30?

- A) 12.2.2.2
- B) 12.2.2.6
- C) 12.2.2.7
- D) none of the above

26. An organization is granted a block; one address is 2.2.2.64/20. The organization needs 10 subnets. What is the subnet prefix length?

- A) /20
- B) /24
- C) /25
- D) none of the above

27. An organization is granted a block; one address is 2.2.2.64/25. If the subnet prefix length is /28, what is the maximum number of subnets?

- A) 2
- B) 4
- C) 8
- D) none of the above

28. An organization is granted a block of classless addresses with the starting address 199.34.76.64/28. How many addresses are granted?

- A) 8
- B) 16
- C) 32
- D) none of the above

29. An organization is granted a block of classless addresses with the starting address 199.34.76.128/29. How many addresses are granted?

- A) 8
- B) 16
- C) 32
- D) none of the above

30. An organization is granted a block of classless addresses with the starting address 199.34.32.0/27. How many addresses are granted?

- A) 8
- B) 16
- C) 32
- D) none of the above

31. What is the default mask for class A in CIDR notation?

- A) /9
- B) /8
- C) /16
- D) none of the above

32. What is the default mask for class B in CIDR notation?

- A) /9
- B) /8
- C) /16
- D) none of the above

33. What is the default mask for class C in CIDR notation?

- A) /24
- B) /8

- C) /16
- D) none of the above

34. In classless addressing, the _____ is another name for the common part of the address range.

- A) suffix
- B) prefix
- C) netid
- D) none of the above

35. In classless addressing, the _____ is the varying part (similar to the hostid).

- A) suffix
- B) prefix
- C) hostid
- D) none of the above

36. In classless addressing, the prefix length defines the _____.

- A) netid
- B) hostid
- C) mask
- D) none of the above

37. In a block, the prefix length is /24; what is the mask?

- A) 255.255.255.0
- B) 255.255.242.0
- C) 255.255.0.0
- D) none of the above

38. In a block, the prefix length is /15; what is the mask?

- A) 255.254.0.0
- B) 255.255.255.0
- C) 255.255.255.128
- D) none of the above

39. In a block, the mask is 255.255.192.0; what is the prefix length?

- A) /20

- B) /28
- C) /18
- D) none of the above

40. An IPv6 address is _____ bits long.

- A) 32
- B) 64
- C) 128
- D) none of the above

41. An IPv6 address consists of _____ bytes (octets);

- A) 4
- B) 8
- C) 16
- D) none of the above

42. To make addresses more readable, IPv6 specifies _____ notation.

- A) dotted decimal
- B) hexadecimal colon
- C) both a and b
- D) none of the above

43. In hexadecimal colon notation, a 128-bit address is divided into _____ sections, each _____ hexadecimal digits in length.

- A) 8: 2
- B) 8: 3
- C) 8: 4
- D) none of the above

44. An IPv6 address can have up to _____ colons.

- A) 8
- B) 7
- C) 4
- D) none of the above

45. An IPv6 address can have up to _____ hexadecimal digits.

- A) 16
- B) 32
- C) 8
- D) none of the above

46. In IPv6, _____ address defines a single computer.

- A) a unicast
- B) a multicast
- C) an anycast
- D) none of the above

47. In IPv6, _____ address defines a group of computers with addresses that have the same prefix.

- A) a unicast
- B) a multicast
- C) an anycast
- D) none of the above

48. In IPv6, _____ address defines a group of computers.

- A) a unicast
- B) a multicast
- C) an anycast
- D) none of the above

49. In IPv6, the _____ prefix defines the purpose of the address.

- A) type
- B) purpose
- C) both a and b
- D) none of the above

50. In IPv6, the _____ address is generally used by a normal host as a unicast address.

- A) provider-based unicast
- B) link local

- C) site local
- D) none of the above

51. In IPv6, a _____ address comprises 80 bits of zero, followed by 16 bits of one, followed by the 32-bit IPv4 address.

- A) link local
- B) site local
- C) mapped
- D) none of the above

52. In IPv6, a _____ address is an address of 96 bits of zero followed by 32 bits of IPv4 address.

- A) link local
- B) site local
- C) mapped
- D) none of the above

53. In IPv6, a _____ address is used if a LAN uses the Internet protocols but is not connected to the Internet for security reasons.

- A) link local
- B) site local
- C) mapped
- D) none of the above

54. In IPv6, a _____ address is used if a site with several networks uses the Internet protocols but is not connected to the Internet for security reasons.

- A) link local
- B) site local
- C) mapped
- D) none of the above